NTPclock Software Documentation
(last updated 20261104-1130)

DESCRITPION OVERVIEW

The NTPclock project, which displays its primary screen as shown in Figure 1 below, is an extension of the original NTPclock by Bruce Hall (W6BH). This particular version has been created in preparation for build-a-thon activities at the VK2MB Manley Warringah Radio Society (MWRS), the K5COW Cowtown Amateur Radio Club (CARC), and the Radio and Electronics Association of Southern Tasmania (REAST). This document describes the capabilities that are provided by the software/firmware loaded into the NTPclock, and also includes instructions for building the source code and loading it into the NTPclock hardware.

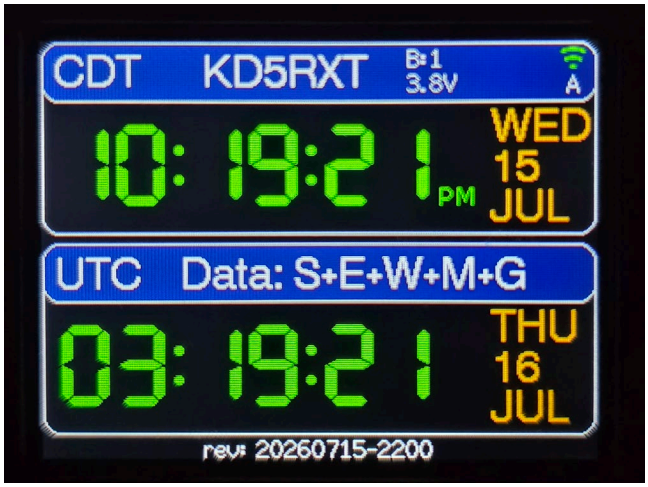


Figure 1. NTPclock Primary Screen

CAPABILITIES OVERVIEW

The firmware loaded into the NTPclock hardware automatically runs at power-up, and provides the following capabilities:

- Simultaneously displays time/date in two formats:
  - UTC/GMT time/date in the bottom panel
  - Local time/date cycling thru the selected time zones in the top panel
- Displays selectable data groups as follows:
  - Solar weather
  - Environmental data
  - Local weather data
  - Maidenhead Grid Square (MLS)
  - GPS data

- Allows the user to easily enable/disable an integrated GPS receiver(optional)
- Selectable Day mode / Night mode display
- Configurable colors
- User-initiated reboot
- Wi-Fi information display
- Metric / Imperial metric unit selection
- Seven fixed display brightness levels, as well as an automatic light-detecting mode
- In addition to some configuration selections that can be made using the two built-in buttons, all configuration selections can be made via the AP-hosted webpage using a browser
- Ability to restore to factory default configuration
- Ability to connect to a local Wi-Fi signal, either automatically or manually

CONSIDERATIONS/LIMITATIONS

The family of ESP32 processors produced by Espressif includes a variety of members with a wide range of capabilities. The NTPclock project was designed to run specifically on an ESP32-S2-mini processor. This particular processor was selected for its low cost, its large number of GPIO pins (allowing connection of a TFT display, a BMP-280 sensor, supply voltage measurement, display dimming, etc.), its ability to host an AP capability supporting a configuration webpage, and the ease with which it can be customized with software/firmware programming.

In spite of all of its advantages, it does also have some limitations to be aware of. First and foremost, the ESP32-S2-mini processor module has a limited amount (320K) of available RAM. With all of the capabilities that this version of the NTPclock now makes available, the utilization of the limited RAM resources requires special considerations. The authors of this version of the NTPclock firmware source have striven to do their very best at managing the utilization of the limited RAM resources to maximize capability, while also working to minimize any potentially undesirable side effects

The ESP32-S2-mini processor contains insufficient RAM to have/leave the GPS enabled while also fetching solar data from the internet. With this in mind, whenever the GPS is activated/enabled, solar data fetching is automatically temporarily suspended. It is important to be aware that, if the solar data selection is re-enabled while the GPS is also enabled, the fetching of solar data will likely fail due to the lack of sufficient RAM required to support successful execution of the secure TLS handshake - the NTPclock may or may not also reboot itself if/when these handshake failures occur. Please see the scenario descriptions later in this document for additional guidance on how to successfully make use of all of the capabilities provided by the NTPclock, while also managing these limitations when utilizing the ESP32-S2-mini processor module.

Taking these limitations into consideration, the current recommendation is to build the NTPclock project utilizing an ESP32-S3-mini processor module, as it is newer and includes even more RAM, allowing the use of all capabilities without any imposed limitations.

INITIALIZATION

Upon initial power-up, the NTPclock reads configuration settings from non-volatile storage (EEPROM) & configures itself using these settings. An initial splash screen is displayed showing, among other things, the revision level of the loaded firmware, as shown in Figure 2 below:

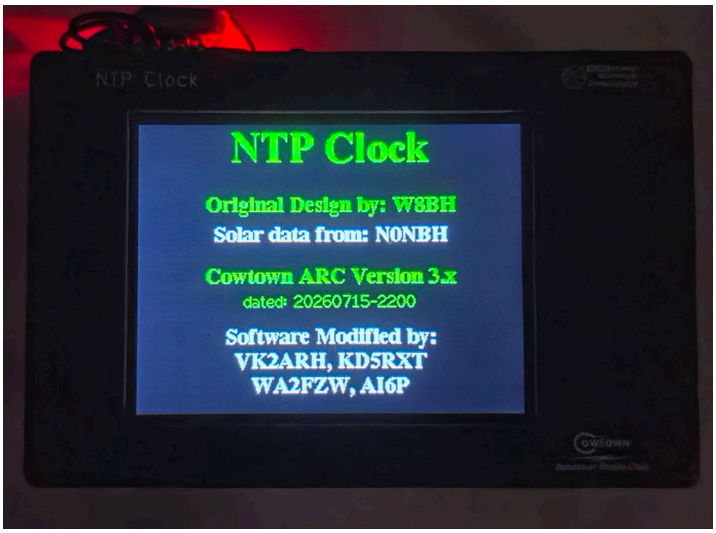


Figure 2. NTPclock Initial Splash Screen

Following display of the initial splash screen, a reminder with the details on how to obtain the (free) API Key required to fetch internet weather data (this is skipped if the API Key has been set) as shown in Figure 3 below:

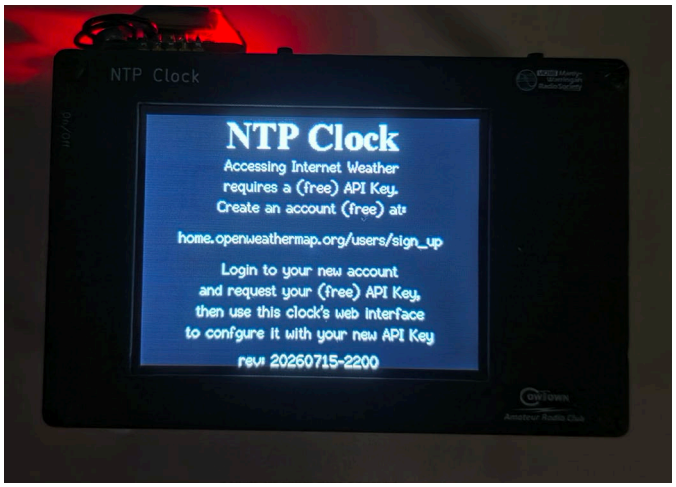


Figure 3. NTPclock API Key Reminder Screen

Following this, the NTPclock will then start up the AP host, followed by an attempt to make a Wi-Fi connection (this is skipped on a newly built NTPclock, as the Wi-Fi configuration will be empty), followed by a display of the key to the button operations as shown in Figure 4 below:

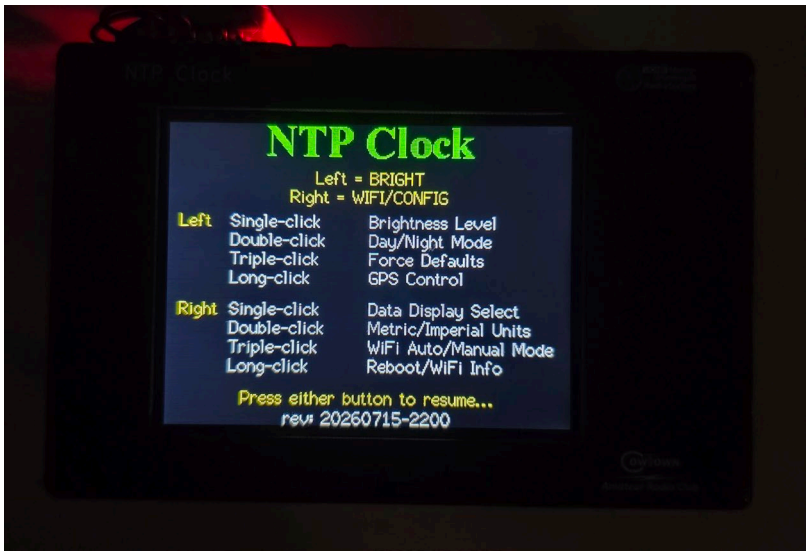


Figure 4. NTPclock Button Operations Screen

NOTE: To return to this display of the key to the button operations at any time, press the buttons as follows (and, as indicated in this display, press either button to dismiss this screen and resume the NTPclock functionality):

1) Press and hold the **WIFI/CONFIG** (right-hand) button

2) Press and release the **BRIGHT** (left-hand) button
3) Release the **WIFI/CONFIG** (right-hand) button

- OR –

1) Press and hold the **BRIGHT** (left-hand) button
2) Press and release the **WIFI/CONFIG** (right-hand) button
3) Release the **BRIGHT** (left-hand) button

When using the CARC NTPclock v4 PCB (20261019), if GPS was enabled the last time the clock was powered off, then the clock will attempt to reactivate and reinitialize the GPS at this point. Finally, the NTPclock will show the as-configured primary clock display as shown earlier in Figure 1.

LOCAL TIME DISPLAY

The NTPclock supports the display of a number of local time zones (and associated dates) in the lower portion of the upper panel, including the following (others can be added if/as necessary - see the description in the DEFAULTS section later in this document for details on how to do this):

- EST/EDT US Eastern time
- CST/CDT US Central time
- MST/MDT US Mountain time
- MST Arizona time (no DST)
- PST/PDT US Pacific Time
- AWST AUS Western time
- ACST/ACDT AUS Central time
- AEST/AEDT AUS Eastern time
- GMT/BST UK time

The format of the displayed time and date (in both upper and lower panels) can be configured with the following options:

- 12/24-hour format for local time
- 12/24-hour format for UTC/GMT time
- Selectable display of a leading zero for local time/hours
- Selectable display of a leading zero for local date/day

- Selectable date above/below month

In addition to the display of local time, the upper panel is also used to display the configurable NTPclock title (up to 8 characters, configurable), the current brightness selection, the status of the GPS (when enabled), the measured voltage level of the V_{bus} supply (for the v2.7a/b PCB, when enabled) or V_{bat} battery voltage (for the CARC NTPclock v4 PCB (20261019) when enabled), the measured voltage level of the 12VDC supply (for the v2.7a/b PCB only, when enabled), the status of the local Wi-Fi connectivity, and whether Wi-Fi reconnection is manual or automatic. A visualization of the available display areas is shown below in Figure 5.

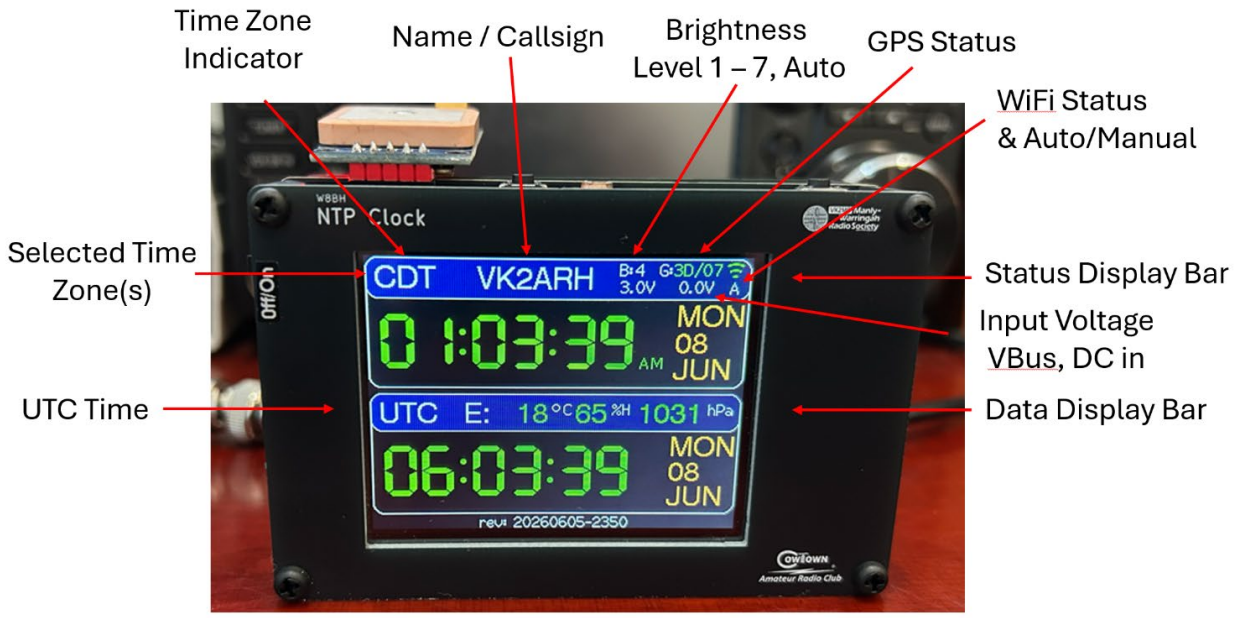


Figure 5. NTPclock Available Display Areas

UTC/GMT TIME DISPLAY

The NTPclock supports the display of UTC/GMT time in the lower panel. UTC/GMT time and data are displayed in the left-hand portion of the lower panel. In addition, the banner portion of the lower panel is used to display the following data (user selectable):

- Solar weather (fetched from the internet, when Wi-Fi connectivity has been established)
    - Solar Flux Index (SFI)
    - A Index (A)
    - K Index (K)
    - Geo Magnetic Fiend (GMF)
    - Signal-to-Noise (S2N)

  - Aurora (AUR)
  - Magnetic Field Direction (BZ)
  - Sunspot Number (SSN)
- Environmental data (using the on-board BME-280/BMP-280 sensor)
  - Temperature
  - Humidity (BME-280 only)
  - Barometric pressure
- Weather data (fetched from the internet, when Wi-Fi connectivity has been established)
  - Displays the following data types
    - Location
    - Current conditions
    - Temperature (with “feels like” temperature as well)
    - Relative humidity
    - Wind speed (with gust speed, as appropriate) and direction
    - Time of sunrise
    - Time of sunset
  - Uses GPS location data to fetch local area weather data
- Maidenhead Grid Square (MLS)
  - Uses GPS location data to calculate grid square data
- GPS data
  - Displays the following data types:
    - Latitude
    - Longitude
    - Altitude (selectable units: feet / meters)

DEFAULTS

All default settings are configurable in the UserSettings.h file (if you care to make changes before building and loading a customized copy of the firmware into the NTPclock hardware). In addition, the list of selectable time zones is also set in this same file. If you plan to use your NTPclock in a time zone

that is not currently included in the existing NTPclock, the list of definitions is easily updated to add a new POSIX Timezone String for the needed time zone as follows (add it wherever desired to reside in the list, paying particular attention to following the proper use of a comma "," to separate definitions):

"STDxDTD,Mm.n.d/hh:mm:ss,Mm.n.d/hh:mm:ss"

Where each field is defined as follows:

x = represents the number of hours that local time differs (a positive number indicates that local time is behind UTC/GMT, and a negative number indicates that local time is ahead of UTC/GMT) from UTC/GMT (e.g. CST6CDT would mean that local time in the US Central Timezone is 6 hours behind UTC/GMT when Daylight Saving Time is not active)

m = represents the month during which time changes, where the first "m" is for the month when the time changes to enter Daylight Saving Time, and the second "m" is for the month when the time changes to discontinue Daylight Saving Time (e.g. M3 indicates that the time changes in the third month)

.n = represents the occurrence of the day in the month on which time changes (e.g. .2 indicates that time changes on the second occurrence of the specified day)

.d = represents the day of the week on which time changes (e.g. .0 indicates that time changes on Sunday)

/hh:mm:ss = represents the time of day at which time changes (e.g. /2:00:00 indicates that time changes at 2:00:00 am)

Detailed example (using Central Standard Time / Central Daylight Time):

"CST6CDT,M3.2.0/2:00:00,M11.1.0/2:00:00"

- Daylight Saving Time is activated in the 3rd month (March), on the second occurrence of Sunday, at 2:00:00 am
- Daylight Saving Time is discontinued in the 11th month (November), on the first occurrence of Sunday, at 2:00:00 am

BUTTON OPERATIONS

The functionality provided by each of the two buttons is as follows (NOTE: see the scenario descriptions later in this document showing examples of how the individual capabilities (controlled by the button functions) might be used):

BRIGHT (left-hand) button:

Single-click (brightness): cycles through and selects from the fixed (1-7) brightness levels, and the automatic light-detecting mode

Double-click (day/night mode): toggles between day mode (configurable colors) and night mode (all red)

Triple-click (force defaults): provides the ability to return all settings of the NTPclock to factory defaults (requires confirmation, allows cancellation, and automatically times out after 20 seconds of no activity/selection)

Long-click (> 2 seconds) (GPS control): provides the ability to either enable/disable the GPS, or to perform a factory reset on the GPS (automatically times out after 20 seconds of no activity/selection)

Extra-long-click (> 8 seconds) (DEBUG): enable DEBUG capabilities, which will then become visible as a DEBUG menu tab at the top of the AP hosted webpage (after a page refresh)

WIFI/CONFIG (right-hand) button:

Single-click (data selection): allows for the selection of which data groups (Solar, Environmental, Weather, Maidenhead Grid Square (MLS), and GPS) will be displayed

Double-click (metric/imperial units): toggles between displaying data in metric units and displaying data in imperial units

Triple-click (Wi-Fi auto/manual): toggles between Wi-Fi automatic (re)connection mode and Wi-Fi manual (re)connection mode

Long-click (> 2 seconds) (reboot/Wi-Fi info): provides the ability to reboot the NTPclock (requires confirmation, and automatically times out after 20 seconds of no activity/selection) or to display a Wi-Fi information screen providing particulars on the current Wi-Fi connectivity

Extra-long-click (> 8 seconds) (Wi-Fi password reset): provides the ability to reset/remove any previously set Wi-Fi configuration password protecting the NETWORK webpage

SHORT-CUTS AND QUICK OPERATIONS

The following short-cuts and/or quick operations are available in the NTPclock firmware (when any of these actions are performed, solar updates will occur within the next 5 seconds, and weather updates will occur within the next 30 seconds, as appropriate):

- To quickly force a solar data update/fetch (if and only if solar data is selected for display), double-click the WIFI/CONFIG (right-hand) button to toggle between Metric units and Imperial units. Use this toggle again to return to the desired units.
- To quickly force a weather data update/fetch (if and only if weather data is selected for display), double-click the WIFI/CONFIG (right-hand) button to toggle between Metric units and Imperial units. Use this toggle to return to the desired units.

- To quickly force a solar data update/fetch (if and only if solar data is selected for display), single-click the WIFI/CONFIG (right-hand) button to change the selected data groups for display to any combination that includes solar data (S).
- To quickly force a weather data update/fetch (if and only if weather data is selected for display), single-click the WIFI/CONFIG (right-hand) button to change the selected data groups for display to any combination that includes weather data (W).
- To quickly force a solar data update/fetch (if and only if solar data is selected for display), from the AP hosted CONFIG webpage, change the selected data groups for display to any combination that includes solar data (S) and activate the Save/Apply button.
- To quickly force a weather data update/fetch (if and only if weather data is selected for display), from the AP hosted CONFIG webpage, change the selected data groups for display to any combination that includes weather data (W) and activate the Save/Apply button.
- To quickly deselect displaying solar data, enable the (optional) GPS unit. The (optional) GPS unit can then subsequently be disabled, unless it is desired to leave it enabled (see and heed the warning elsewhere in this document about not enabling the display of solar data when the (optional) GPS unit is enabled/active).

SCENARIO DESCRIPTIONS

The following scenario descriptions can be used to better understand the variety of operating conditions and/or location that the NTPclock is targeted to support.

In the shack:

When operating the NTPclock in the home shack, operating is generally much easier due to the availability of stable power and Wi-Fi connectivity.

Assumptions:

- Stable power is readily available (USB 5VDC and/or nominal 12VDC)
- Wi-Fi connectivity is readily available
- Your position/location is generally fixed (not moving)
- The GPS unit is installed (optional)

Operations:

In this case, configuration of the NTPclock would start with setting up for Wi-Fi connectivity. Initially, the GPS would be enabled long enough to provide/capture good position information. Once a successful 3D fix is reported and displayed, the GPS would then be disabled (causing the current position data to be captured and saved for continued use). With the GPS disabled & Wi-Fi successfully connected, the display of solar data can then be enabled. Additionally, the display of Environmental and MLS data can also be enabled, if desired.

Considerations:

In the event that the optional GPS unit is not installed, the correct position information can also be manually configured from the AP hosted webpage.

Night mode and brightness selections can be used for operations under low-light conditions and to suit individual preferences.

In the field (e.g. POTA):

When operating the NTPclock in the field, available power options and Wi-Fi may vary, so the options need to be customized to suit the individual situation.

Assumptions:

- Stable power may or may not be available (USB 5VDC and/or nominal 12VDC)
- Wi-Fi connectivity may or may not be available
- Your position/location may or may not necessarily be fixed (not moving)
- The GPS unit may or may not be installed (optional)

Operations:

In this case, configuration of the NTPclock would start with setting up for Wi-Fi connectivity, if it is available. Initially, the GPS would be enabled long enough to provide/capture good position information. Once a successful 3D fix is reported and displayed, the GPS would then be disabled (causing the current position data to be captured and saved for continued use). With the GPS disabled & Wi-Fi successfully connected, the display of solar data can then be enabled. Additionally, the display of Environmental and MLS data can also be enabled, if desired.

Considerations:

In the event that Wi-Fi connectivity is not available, then neither solar data nor weather data would be enabled, as Wi-Fi is required to fetch both of those data sources. In that case, only Environmental, MLS, and GPS data would be available for display. In the event that the optional GPS unit is not installed, the correct position information can also be manually configured from the AP hosted webpage. If the optional GPS unit is installed, then it can be removed once a successful 3D fix is obtained. This would help with extending the duration of operations on battery power in the event that stable power is not otherwise available.

NOTE: The NTPclock depends upon an external time source for proper operations. It will not operate correctly without either W-Fi connectivity or a GPS signal. Wi-Fi is used as the primary time source (via NTP), and a GPS based time signal can also be used to maintain clock accuracy.

Accessing the NETWORK webpage on an NTPclock on which password protection has been added, but forgotten

An optional username + password can be set to protect the (potentially sensitive) Wi-Fi configuration information set and held in the NETWORK webpage. If this password has been forgotten, it can be reset/removed to restore access.

Assumptions:

- The optional username + password has previously been set on the NETWORK webpage

Operations:

In this case, simply executing an Extra-long-click (> 8 seconds) on the WIFI/CONFIG (right-hand) button will reset/remove the forgotten password. Note, the username, if configured, remains unchanged.

Considerations:

You should be aware that anyone with physical access to your particular NTPclock can freely and easily execute this recovery procedure to remove any applied password, thus defeating the protection.

Returning the NTPclock to a known good configuration:

When operating the NTPclock, if there is ever any concern for a set of configuration selections and the potential undesirable effects of that particular combination of selections, the NTPclock does include the capability to quickly and easily return to a factory fresh configuration.

Assumptions:

- Incorrect operations of the NTPclock are suspected

Operations:

In this case, the restoration of the default configuration of the NTPclock can be quickly accomplished by executing a triple-click on the BRIGHT (left-hand) button. Following the displayed prompts, subsequently single-clicking the WIFI/CONFIG (right-hand) button will initiate the default restoration process, and subsequently single-clicking the BRIGHT (left-hand) button will cancel the default restoration process. Allowing for the 20-second timeout to expire without activating either button will also cancel the default restoration process.

Considerations:

All configuration changes (Wi-Fi definitions, time/date display configurations, time zone selections, etc.) that might have been made will be lost, as they are replaced with the

restoration of the default configuration. If an API Key has been entered, allowing for fetching internet weather data, this API key is retained.

WEB CONFIGURATION

Many aspects of the NTPclock can be configured thru its AP hosted webpage. As a reminder, the AP host is always running on the NTPclock, regardless of whether a successful Wi-Fi connection has been made or not (but be aware that, anytime the NTPclock attempts to (re)initialize the Wi-Fi connectivity, the AP host may be temporarily interrupted while network the connection is updated). Once the NTPclock is up & running, you can access its AP hosted webpage interface by connecting another device to the Wi-Fi AP titled "NTPCLOCK_AP" (which is the default AP name used by the NTPclock, unless you have previously changed it to something different), then opening a browser and directing it to 10.1.1.1 (alternately, if your NTPclock has already successfully connected to one of its six Wi-Fi definitions, you can connect direct your browser to the IP address supplied by the Wi-Fi router that your NTPclock is connected to).

STATUS webpage

The STATUS webpage can be accessed to get insight into the current Wi-Fi and AP connectivity. The following status items are displayed:

Version - shows the firmware version that is currently executing in the NTPclock

AP - shows both the IP address and the hostname of the AP host used to access the AP hosted webpages

Wi-Fi - shows the name of the Wi-Fi router that the NTPclock is currently connected to (or it shows "(not connected)" in the event that a successful Wi-Fi connection has not (yet) been established to an external Wi-Fi router)

Wi-Fi IP - shows the IP address that the Wi-Fi router has assigned to the NTPclock when a successful Wi-Fi connection has been established (or it shows "0.0.0.0" in the event that a successful Wi-Fi connection has not (yet) been established to an external Wi-Fi router)

Channel - shows the Wii channel on which the AP host is currently operating

MAC - shows the network MAC address of the ESP32-S2-mini in this particular NTPclock

NETWORK webpage

From the NETWORK webpage, the following items can be configured:

Activating the "Hide Passwords checkbox" and subsequently activating the Save/Apply button will cause all passwords on the NETWORK webpage to be displayed as a string of asterisks. Deactivating the "Hide Passwords checkbox" and subsequently activating the Save/Apply button will cause all passwords to be fully viewable as plaintext.

The AP name for this particular NTPclock can be configured (allowing a unique distinction to be made among potentially multiple NTPclocks that may be operating within the same local area).

The Wi-Fi channel over which the AP host of this particular NTPclock will be accessible.

An optional username + password can be set to protect access to the (potentially sensitive) Wi-Fi configurations on the NETWORK webpage

Up to six network definitions (SSID + password) can be entered. Note, upon power-up, the NTPclock will cycle thru the list of six network definitions, skipping any blank entries, and attempt to make a successful Wi-Fi connection using these definitions. Similarly, if the list of six network definitions is updated/saved, and a successful Wi-Fi connection has not previously been established, the NTPclock will once again cycle thru the list of six network definitions, skipping any blank entries, and attempt to make a successful Wi-Fi connection using these updated definitions.

Activating the Save/Apply button at the bottom of the NETWORK webpage activates/applies any changes that may have been made.

COLORS webpage

From the COLORS webpage, the following items can be configured:

Screen Background - sets the color of the background behind the entire screen.

Clock Digits - self explanatory

Date - self explanatory

Label Foreground - sets the color of the foreground characters in the top portion of each of the upper and lower time panels

Label Background - sets the color of the background behind the characters in the top portion of each of the upper and lower time panels

Edge - sets the color of the border around each of the upper and lower time panels

Normal Readings - sets the color of any status indications that are in a “normal” (not “medium”, “high”, or “warning”) state

Medium Readings - sets the color of any status indications that are in a “medium” (not “normal”, “high”, or “warning”) state

High Readings - sets the color of any status indications that are in a “high” (not “normal”, “medium”, or “warning”) state

Lost Wi-Fi - sets the color of the status indication when Wi-Fi connectivity was previously established and has subsequently been lost

Warnings - sets the color of any status indications that serve as a warning of any unusual condition(s)

Activating the "Return All Colors To Defaults" checkbox, then subsequently activating the Save/Apply button at the bottom of the COLORS webpage will reconfigure the display with the default color scheme. Deactivating the "Return All Colors To Defaults" checkbox has no effect.

Activating the Save/Apply button at the bottom of the COLORS webpage activates/applies any changes that may have been made.

CONFIG webpage

From the CONFIG webpage, the following items can be configured:

Data Display Selections

Show Solar Data - enable/disable the display of solar data from the internet.

Show Environmental Data - enable/disable the display of environmental data from the (optional) BMP-280/BME-280 sensor

Show Weather Data - enable/disable the display of weather data from the internet.

Show Maidenhead Grid Square (MLS) Data - enable/disable the display of the calculated MLS grid square

Show GPS Data - enable/disable the display of GPS data (either live data from the (optional) GPS unit, or manual data entered further down the CONFIG webpage)

Banner Title

Banner Title - sets the banner text (up to 8 characters in length) displayed at the top of the upper time panel

GPS Mode

GPS Mode - selects whether the (optional) GPS unit is enabled ("GPS Active Mode") or disabled ("GPS Manual Mode"). Selecting "GPS Manual Mode", and subsequently activating the Save/Apply button at the bottom of the CONFIG webpage makes additional fields visible, into which GPS latitude, GPS longitude, and QTH altitude information can be entered manually.

Display GPS Altitude Units

GPS Altitude Units - selects whether the GPS altitude is displayed in feet or meters (independent of the current metric/imperial selection).

QTH Altitude in Feet

Altitude at Current QTH – allows manual entry of the altitude in feet at the current QTH.

Lat/Lon Decimal Places

Max Number of Decimal Places To Show In Lat/Lon For Weather Display (1-6) - sets the number of decimal places to show (valid range: from 1 to 6 places - the default value is 6 places)

Time zone Display Interval

Number of Seconds To Display Each Active Time zone (1-30) - sets the duration of displaying any time zones selected for display (valid range: 1 to 30 seconds – the default value is 3 seconds)

Time zones to Display

Display the PST/PDT Timezone - US Pacific Time
Display the MST Timezone - Arizona time (no DST)
Display the MST/MDT Timezone - US Mountain time
Display the CST/CDT Timezone - US Central time
Display the EST/DTS Timezone - US Eastern time
Display the GMT/BST Timezone - UK time
Display the AWST Timezone - AUS Western time
Display the ACST/ACDT Timezone - AUS Central time
Display the AEST/AEDT Timezone - AUS Eastern time

NOTE: If any additional timezone definitions have been added to the UserSettings.h file, these additional entries will also be presented for selection in this section as well.

Time & Date Display Format

Show AM/PM Mode - selecting “Don’t Show AM/PM” (no AM/PM display), “Show Small AM/PM” (show “AM” or “PM” horizontally next to local time), or “Show Large AM/PM” (show “AM” or “PM” vertically next to local time), then subsequently activating the Save/Apply button at the bottom of the CONFIG webpage will cause the selected AM/PM mode to be displayed.

Display Local Time In 24 Hour Format - activating this checkbox, then subsequently activating the Save/Apply button at the bottom of the CONFIG webpage will cause the local time in the upper panel to be displayed in 24-hour format, otherwise it will be displayed in 12-hour format.

Display UTC Time In 24 Hour Format - activating this checkbox, then subsequently activating the Save/Apply button at the bottom of the CONFIG webpage will cause the UTC/GMT time in the lower panel to be displayed in 24-hour format, otherwise it will be displayed in 12-hour format.

Time (hour) Is Displayed With A Leading Zero - activating this checkbox is activated, then subsequently activating the Save/Apply button at the bottom of the CONFIG webpage will cause the time display to include a leading zero on the hours, otherwise no leading zero will be shown.

Date (day) Is Displayed with a Leading Zero - activating this checkbox, then subsequently activating the Save/Apply button at the bottom of the CONFIG webpage will cause the date display to include a leading zero on the day, otherwise no leading zero will be shown.

Date (day) Is Displayed Above The Month - activating this checkbox, then subsequently activating the Save/Apply button at the bottom of the CONFIG webpage will cause the date display of the day to be drawn above the month, otherwise the month will be drawn above the day.

[The capability described in the Voltage Measurement Display section is only available when using the NTPclock v2.7a/b PCB. See the following section for the available capability description when using the CARC NTPclock v4 PCB (20261019).]

Voltage Measurement Display

5VDC Supply Measurement Is Displayed - activating this checkbox, then subsequently activating the Save/Apply button at the bottom of the CONFIG webpage will cause the 5VDC measurement to be shown in the upper panel, otherwise the 5VDC measurement will not be shown.

5VDC Supply Measurement Correction/Calibration Factor (Default Value = 1.0) - this correction/calibration value will be used as a multiplier when displaying the measured value of the 5VDC supply. The actual measurement is taken at the processor V_{bus} pin and reflects the voltage being supplied to the processor. This value will generally be slightly lower than the USB-C input voltage (5V) as the supply will have passed through a protection diode and switch before reaching the processor. Similarly, when powering the NTPclock using a LiPo battery, this voltage will reflect the power supplied from the LiPo battery to the processor. This voltage reading provides a good indication of battery health. The needed correction/calibration factor can be determined as follows:

> Ensure that the “5VDC Supply Measurement Correction/Calibration Factor” field contains the value of “1.0”. Measure the actual physical value of the V_{bus} supply using a multimeter (call this V_{actual}). Reference Figure 6 below which shows readily accessible locations for measuring V_{bus} on the rear of the v2.7a/b PCB on a fully built assembly. Observe the displayed value of the ‘5VDC’ supply indication in the upper panel of the display (call this $V_{display}$). Calculate the new correction/calibration value by dividing the value of V_{actual} by the value of $V_{display}$. Enter this correction/calibration value in the “5VDC Supply Measurement Correction/Calibration Factor” field and subsequently activate the Save/Apply button at the bottom of the CONFIG webpage.

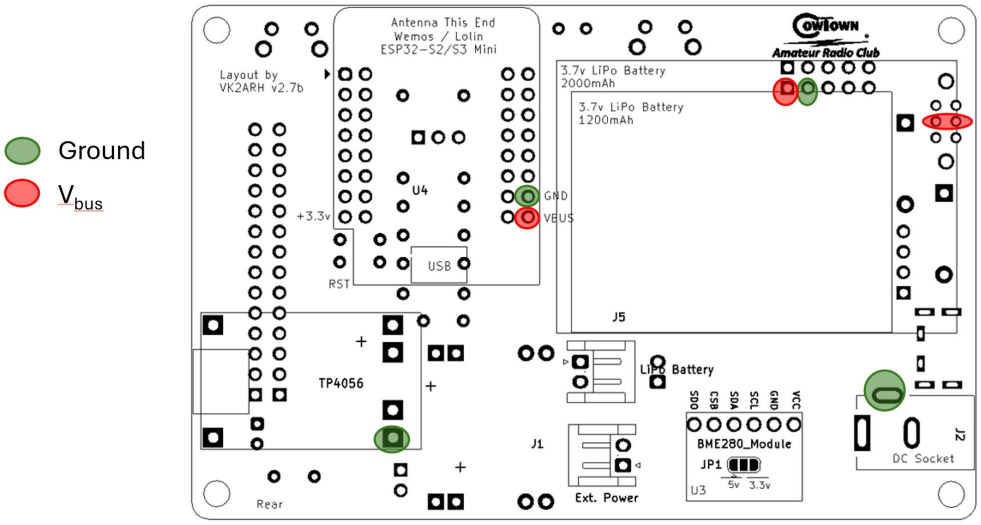


Figure 6. NTPclock v2.7a/b PCB Locations for Measuring V_{bus}

5VDC Supply Measurement Is Displayed - activating this checkbox, then subsequently activating the Save/Apply button at the bottom of the CONFIG webpage will cause the V_{bus} measurement to be shown in the upper panel, otherwise the 5VDC measurement will not be shown.

12VDC Supply Measurement Correction/Calibration Factor (Default Value = 1.0) - this correction/calibration value will be used as a multiplier when displaying the measured value of the 12VDC (DC_{in}) supply. The needed correction/calibration factor can be determined as follows:

> Ensure that the "12VDC Supply Measurement Correction/Calibration Factor" field contains the value of "1.0". Measure the actual physical value of the DC_{in} supply using a multimeter (call this V_{actual}). Reference Figure 7 below which shows readily accessible locations for measuring DC_{in} on the rear of the PCB on a fully built assembly. Observe the displayed value of the 12VDC supply indication in the upper panel of the display (call this $V_{display}$). Calculate the new correction/calibration value by dividing the value of V_{actual} by the value of $V_{display}$. Enter this correction/calibration value in the "12VDC Supply Measurement Correction/Calibration Factor" field and subsequently activate the Save/Apply button at the bottom of the CONFIG webpage.

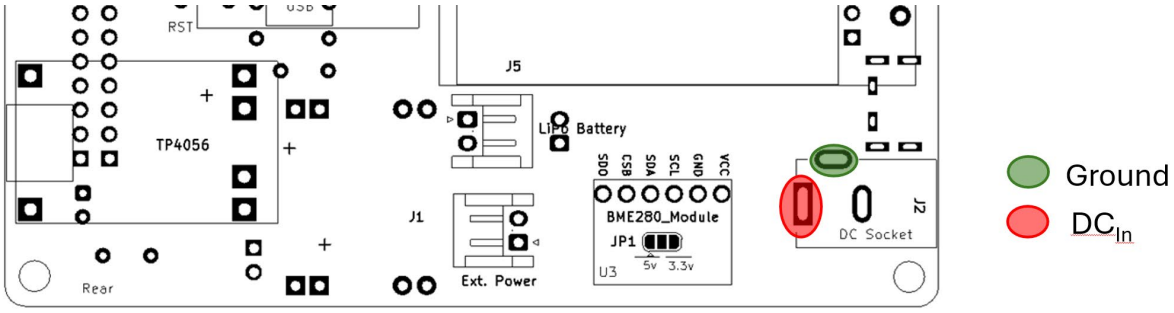


Figure 7. NTPclock v2.7a/b PCB Locations for Measuring DC_{in}

[The capability described in the Battery Voltage Display section is only available when using the CARC NTPclock v4 PCB (20261019). See the previous section for the available capability description when using the NTPclock v2.7a/b PCB.]

Battery Voltage Display

Battery Voltage Measurement Is Displayed - activating this checkbox, then subsequently activating the Save/Apply button at the bottom of the CONFIG webpage will cause the battery voltage measurement to be shown in the upper panel, otherwise the battery voltage measurement will not be shown.

Battery Voltage Measurement Correction/Calibration Factor (Default Value = 1.0) - this correction/calibration value will be used as a multiplier when displaying the measured value of the battery voltage. The actual measurement is taken at the processor V_{bat} pin and reflects the battery voltage being supplied to the battery input of the Adafruit PowerBoost 1000C battery management module. This voltage reading provides a good indication of battery health. The needed correction/calibration factor can be determined as follows:

> Ensure that the “Battery Voltage Measurement Correction/Calibration Factor” field contains the value of “1.0”. Measure the actual physical value of the V_{bat} supply using a multimeter (call this V_{actual}). Reference Figure 8 below which shows readily accessible locations for measuring V_{bat} on the front of the CARC NTPclock v4 PCB (20261019) on a fully built assembly. Observe the displayed value of the battery voltage indication in the upper panel of the display (call this $V_{display}$). Calculate the new correction/calibration value by dividing the value of V_{actual} by the value of $V_{display}$. Enter this correction/calibration value in the “Battery Voltage Measurement Correction/Calibration Factor” field and subsequently activate the Save/Apply button at the bottom of the CONFIG webpage.

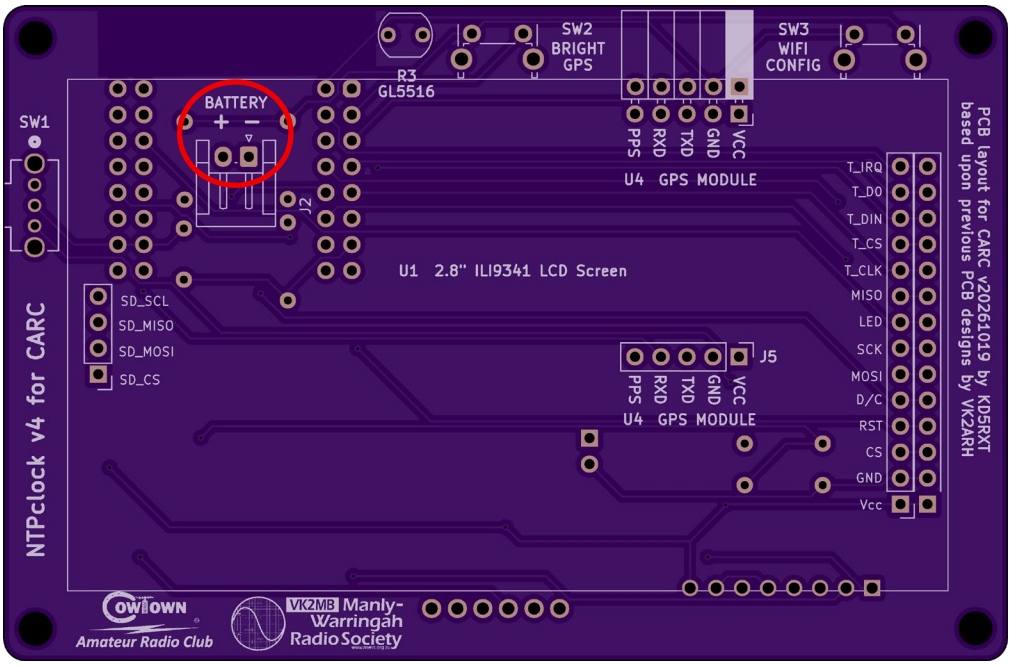


Figure 8. CARC NTPclock v4 PCB (20261019) Locations for Measuring V_{bat}

Printed Time Selection

Printed Time - selecting “No Printed Time” (no time printed), “Print UTC Time” (print only UTC/GMT), “Print Local Time” (print only local time), or “Print UTC & Local Time” (print both UTC/GMT time and local time), then subsequently activating the Save/Apply button at the bottom of the CONFIG webpage will cause the selected time(s) to be printed to the Arduino IDE Serial Monitor.

Blink Time Update For Visual Syncing

Blink the (RGB) LED With Each Time Update - activating this checkbox, then subsequently activating the Save/Apply button at the bottom of the CONFIG webpage will cause the (RGB) LED on the processor to blink each time that the time display is updated. This indication is a visual debugging capability that can be used as a comparison to a synchronized time source (e.g. GPS PPS indicator) as an indication of how close the NTPclock time is updated to the nearest one-second transition.

Show Selected Data Header Between Data Groups

Show Selected Data Header Between Data Groups - activating this checkbox, then subsequently activating the Save/Apply button at the bottom of the CONFIG webpage will cause the header showing which data groups have been activated/selected to be displayed the first data value of each data group is displayed.

Temperature Offset

Temperature Offset (Entered In Degrees Celsius) - any value entered into this field will be added (i.e. to cause a reduction in the temperature, enter a negative number) to the temperature measured/reported by the (optional) BMP-280/BME-280 sensor. This can be used to cancel out any localized heating caused by other components that may near enough to the BMP-280/BME-280 to cause an undesirable thermal influence.

Humidity Offset

Humidity Offset (Entered In Percent) - any value entered into this field will be added (i.e. to cause a reduction in the humidity, enter a negative number) to the humidity measured/reported by the (optional) BMP-280/BME-280 sensor. This can be used to cancel out any localized change in the humidity caused by other components that may near enough to the BMP-280/BME-280 to cause an undesirable influence.

Weather Fetch Interval

Number of Seconds Between Weather Fetches From The Internet (10-3600) - the value entered into this field will be used to set the number of seconds that will be allowed to expire between network calls to the internet weather server (valid range: 10 seconds to 3600 seconds, or 60 minutes - the default value is 3600 seconds). Since weather data only changes every 10 minutes for a location that has not changed, it is recommended that you set this value to no less than 600 seconds (10 minutes) in order to preserve the number of 'free' API calls you are making to the openweathermap.org server. Decreasing the weather fetch interval to less than this value will not deliver any different result for a given location.

Personal API Key From openweathermap.org

Weather API Key - the key entered into this field must be obtained from home.openweathermap.org/users/sign_up and is unique to each user and/or each NTPclock. Without this API Key, internet weather cannot be fetched.

SCAN webpage

Selecting the SCAN webpage will cause the NTPclock to display a list of 2.4GHz APs that are detected within local range. The following information is displayed for each detected 2.4GHz AP: SSID, RSSI (signal strength), CH (channel), and Cipher (the type of Wi-Fi encryption in use). An automatic update/refresh on the list of 2.4GHz APs that are detected is continuously executed as long as the SCAN webpage is selected/displayed. Note that there may be a slight pause when entering or exiting the SCAN webpage (to allow an in-progress scan to complete).

DEBUG webpage

The selections on the DEBUG webpage are for geeks only, and can be used to monitor a variety of details on the running NTPclock !! In order to make the DEBUG page available for use, DEBUG mode must first be enabled on the NTPclock by executing an Extra-long-click (> 8 seconds) on the BRIGHT (left-hand) button. If any of the AP hosted webpages are open when DEBUG mode is enabled, then a webpage refresh must be executed in the browser (e.g. ALT-F5) in order to subsequently access the DEBUG webpage.

Each of the selections on the DEBUG webpage should be self-explanatory.

FIRMWARE BUILD INSTRUCTIONS

The source for the current version for the firmware which can be loaded into the NTPclock is found at the following location:

https://github.com/mjculross/NTPclock-for-MWRS-CARC/

The firmware for the NTPclock is built using the Arduino IDE (both the older 1.8.19 version and the newer 2.3.10 version can be used to successfully build and load the firmware into the NTPclock hardware).

NOTE: Be sure to select the correct target build in tft_setup.h (default for MWRS using ESP32-S2-mini: (all other selections commented out), CARC using ESP32-S3-mini: "CARC_V4_ESP32S3-MINI" uncommented, or MWRS using ESP32-S3-mini: "MWRS_ESP32S3_MINI" uncommented).

NOTE: Be sure to select the correct processor in the Arduino IDE, using the choices detailed below as your guide.

Details for the Arduino IDE setup are given below:

```
IDE Setup:  = = = = = = = = = = = = = = = = = = = =
                        For the ESP32-S3-mini:
                                -or-
            For the CARC (using the ESP32-S3-mini):
            = = = = = = = = = = = = = = = = = = = =
            In the Arduino IDE, add the ESP32 board support package by
            opening File > Preferences, then add:
               "https://espressif.github.io/arduino-esp32/package_esp32_index.json:
```

(without the quotes) to the "Additional Boards Manager URLs" (separate multiple board support definitions with a comma ",")

In the Arduino IDE, go to Tools > Board:, then select "Boards Manager..."

In the Boards Manager, enter "ESP32" into the search bar & install the "esp32" package (currently version 3.3.10)

In the Arduino IDE, go to Tools > Board: > , then select:

Board: "LOLIN S3 Mini"

Change/verify the following settings for that board type:

Upload Speed: "921600"

USB Mode: "Hardware CDC and JTAG:

USB CDC On Boot: "Enabled"

USB Firmware MSC On Boot: "Disabled"

USB DFU On Boot: "Disabled"

Upload Mode: "UART0 / Hardware CDC"

CPU Frequency: "240MHz (WiFi)"

Flash Mode: "QIO 80MHz"

Partition Scheme: "Huge App (3MB No OTA/1MB SPIFFS)"

Core Debug Level: "None"

Arduino Runs On: "Core 1"

Events Run On: "Core 1"

Erase All Flash Before Sketch Upload: "Disabled"

```
= = = = = = = = = = =
For the ESP32-S2-mini:
= = = = = = = = = = =
In the Arduino IDE, add the ESP32 board support package by
opening File > Preferences, then add:
   "https://espressif.github.io/arduino-esp32/package_esp32_index.json:
(without the quotes) to the "Additional Boards Manager URLs" (separate multiple
board support definitions with a comma ",")
In the Arduino IDE, go to Tools > Board:, then select "Boards Manager..."
In the Boards Manager, enter "ESP32" into the search bar & install the "esp32"
package (currently version 3.3.10)
In the Arduino IDE, go to Tools > Board: > , then select:
   Board: "LOLIN S2 Mini"
Change/verify the following settings for that board type:
   USB CDC On Boot: "Enabled"
   USB Firmware MSC On Boot: "Disabled"
   USB DFU On Boot: "Disabled"
   Partition Scheme: "Huge App (3MB No OTA/1MB SPIFFS)"
   Core Debug Level: "None"
   Erase All Flash Before Sketch Upload: "Disabled"
```

Libraries: **The following libraries must be loaded (typically, by using the Arduino IDE Library Manager):**

TFT_eSPI:	**by Bodmer,**	**version 2.5.43**
ezTime:	**by Rop Gonggrijp,**	**version 0.8.3**
ArduinoJson:	**by Benoit Blanchon,**	**version 7.4.3**
Button2	**by Lennart Hennigs**	**version 2.7.0**
ESPAsyncWebServer:	**by ESP32Async,**	**version 3.12.1**
AsyncTCP	**by ESP32Async**	**version 3.5.0**
TinyGPSPlus:	**by Mikal Hart,**	**version 1.0.3**
SparkFun u-blox GNSS Arduino Library:	**by SparkFun Electronics,**	**version 2.2.29**
EspSoftwareSerial	**by Dirk Kaar**	**version 8.1.0**
Preferences:	**by Volodymyr Shymanskyy,**	**version 2.4.0**
Adafruit BME280 Library (install all):	**by Adafruit,**	**version 2.3.0**
Adafruit BMP280 Library (install all):	**by Adafruit,**	**version 3.0.0**

PROGRAMMING THE ESP32:

NOTE: When programming the ESP32 (any variety) for the very first time, the processor may need to be put into "programming" mode manually as follows (and as shown, using the ESP32-S2-mini as an example, in Figure 8 below):

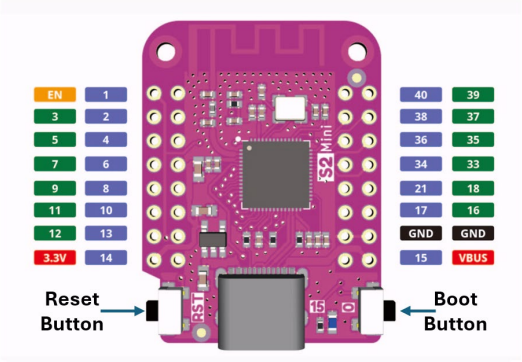


Figure 8. ESP32 programming buttons

1) Press and hold the button on the processor module labeled **0** or **BOOT**
2) Press and release the button on the processor module labeled **RST** or **RESET**
3) Release button **0**

Select the correct serial/COM (the correct serial/COM port can be verified using the Windows Device Manager & watching for which COM port number shows up when the processor module is plugged into a USB port) in the Arduino IDE under **Tools > Port**, then click the **Upload** arrow button. For an ESP32 that was put into programming mode manually, once the programming completes, the **RST button** must be pressed once to reboot the processor module and launch the newly programmed firmware. For subsequent firmware uploads, this manual process is not required.